Chapter 14

Reflections on Cosmology

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umans throughout history have had a passionate drive to understand the origin of the universe. There is, perhaps, no single question that so transcends cultural and temporal divides, inspiring the imagination of our ancient forebears as well as the research of the modern cosmologist. At a deep level, there is a collective longing for an explanation of why there is a universe, how it has come to take the form we witness, and for the rationale—the principle—that drives its evolution. The astounding thing is that humanity has now come to a point where a framework is emerging for answering some of these questions scientifically.

The currently accepted scientific theory of creation declares that the universe experienced the most extreme of conditions—enormous energy, temperature, and density—during its earliest moments. These conditions, as is by now familiar, require that both quantum mechanics and gravity be taken into account, and hence the birth of the universe provides a profound arena for exercising the insights of superstring theory. We will discuss these nascent insights shortly, but first, we briefly recount the pre-string theory cosmological story, which is often referred to as the standard model of cosmology.

The Standard Model of Cosmology

The modern theory of cosmic origins dates from the decade and a half after Einstein's completion of general relativity. Although Einstein refused to take his own theory at face value and accept that it implies that the universe is neither eternal nor static, Alexander Friedmann did. And as we discussed in Chapter 3, Friedmann found what is now known as the big bang solution to Einstein's equations—a solution that declares that the universe violently emerged from a state of infinite compression, and is currently in the expanding aftermath of that primeval explosion. So certain was Einstein that such time-varying solutions were not a result of his theory that he published a short article claiming to have found a fatal flaw in Friedmann's work. Some eight months later, however, Friedmann succeeded in convincing Einstein that there was, in fact, no flaw; Einstein publicly but curtly